

Ali Asgarov

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RESEARCH INTERESTS

I work on computer vision and natural language processing with a focus on multimodal learning, including vision-language reasoning, cross-modal retrieval, and structured information extraction.

EDUCATION

Virginia Tech, PhD in Computer Science, GPA: 4.0/4.0 Aug 2024 – Present
 Advisor: Dr. Chris Thomas
 Coursework: Multimodal Vision, Natural Language Processing, Statistics in Research.
George Washington University, MSc in Computer Science, GPA: 3.76/4.0 Aug 2022 – Dec 2023
 Advisors: Dr. Rebecca Hwa and Dr. Samir Rustamov
 Coursework: Machine Learning, Neural Networks and Deep Learning, Cloud Computing, AI, Big Data and Analytics.
Baku Higher Oil School, BEng in Automation, GPA: 91/100 Aug 2017 – May 2022
 Advisor: Dr. Ali Parsayan
 Coursework: Calculus I–III, Numerical Methods, Algorithms and Data Structures I–III, Image Pattern Recognition.

EXPERIENCE

PhD Applied Scientist Intern, Amazon WA, USA | April 2026 – Present
 Working on multimodal LLMs for Amazon's global returns platform, building agentic conversational systems and vision language models to improve return reasoning, damage assessment, and large scale decision making.
Machine Learning Engineer, Polygraf AI Texas, USA | Dec 2023 – Aug 2024
 Built multimodal deepfake detection models for text, image, audio, and video by fine-tuning transformer-based models using PEFT methods (LoRA, QLoRA, etc). Delivered 92% F1 and 97% recall on high risk content with under 100 ms inference for 25k users.
Machine Learning Engineer, Revenue AI Amsterdam, Netherlands | Feb 2022 – Aug 2022
 Built ML models and data pipelines for anomaly detection, classification, forecasting, and video summarization on Azure ML and Synapse. Led delivery of ML solutions and pipelines for partners including PepsiCo, P&G, and MuSigma.
Machine Learning Engineer, Voiceloft AI Delaware, USA | Jan 2021 – Feb 2022
 Developed multilingual ASR for low resource English accents, achieving 93% accuracy and supporting 30k users at 99.8% uptime.

PUBLICATIONS

NEST Narrative Event Structures in Time for Long Video Understanding Feb 2026
 A. Asgarov, K. Narasimhan, N.H. Sarker, S. Mallampati, A. Sivakumar, C. Thomas | **Accepted**, EMNLP 2026 Main paper ☞
 Lenses: Toward Polysemous Vision–Language Understanding Feb 2026
 Hani Alomari, A. Asgarov, Chris Thomas | **Accepted**, CVPR 2026
 Immunizing Models Against Harmful Long-Horizon Fine-Tuning via Contractive Optimization Dynamics Feb 2026
 N.H.Sarker, Z.I.A. Hakim, A. Asgarov, Chia-Wei Tang, et al. | **Accepted**, CVPR 2026
 Benchmarking and Mitigating MCQA Selection Bias of Large Vision Language Models Sep 2025
 M. Atabuzzaman, A. Asgarov, C. Thomas | **Accepted**, EMNLP 2025 Main paper ☞
 SIGMA Search Augmented On Demand Knowledge Integration for Agentic Mathematical Reasoning Aug 2025
 A. Asgarov, U. Suleymanov, A. Khatri | **Accepted**, AAAI 2026 LMReasoning paper ☞
 Advancing Public Safety Video Sensemaking with Ground Robot Footage May 2025
 P. Zhou, A. Asgarov, A. Hussain et al. | **Accepted**, CHI 2026 paper ☞
 ENTER Event Based Interpretable Reasoning for VideoQA Oct 2024
 H. Ayyubi, J. Liu, A. Asgarov, Z.I.A. Hakim et al. | **Accepted, Spotlight**, NeurIPS 2024 MAR paper ☞

TECHNICAL SKILLS

Languages: Python, C/ C++, SQL, JavaScript, R, MATLAB
Libraries: HuggingFace, OpenCV, Pandas, NumPy, Scikit-learn, NLTK, SpaCy
Frameworks: PyTorch, TensorFlow, FastAPI, Docker, Spark, Hadoop
Cloud: AWS, Azure, GCP

PATENTS

US Patent (Granted). 4883 5431 3873 2024 ☞ System and Method for Identifying and Determining a Content Source.

SERVICE, TEACHING & LANGUAGES

Teaching: CS3114 Data Structures and Algorithms (Spring 2025) and CS5644 ML with Big Data (Fall 2024)
Reviewing: ACM TIST 2025, EMNLP 2025, WACV 2026, CVPR 2026, NeurIPS 2026, EMNLP 2026
Languages: English (fluent), Turkish (fluent), Azerbaijani (native)

HONORS AND AWARDS

2025 Pratt Fellowship, Virginia Tech ☞
2023 State Scholarship for Azerbaijani Youth Abroad ☞
2020–2022, 1st Place (3×) National AI Competition ☞
2020, Top 15 World Robot Olympiad, Hungary ☞
2019, Gold Medal, National Robot Olympiad, Azerbaijan ☞
2018 Presidential Scholarship ☞
2017 University Entrance Exam 690/700
2017 High School Graduation with Gold Medal (Top 0.1%)